

Universal Learning Pattern Recognition Strategy

User Settings Reference Guide

PineScript V6 - Chart Overlay Strategy

This document provides a complete reference for every user-configurable setting in the ULPR Strategy. Settings are organized into seven groups (A through G) matching the Settings/Inputs tab in TradingView. Each setting includes its type, default value, valid range, and a description of its effect on strategy behavior. Practical examples demonstrate common configuration scenarios for different asset classes and timeframes.

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Section A - Trend Core

The Trend Core determines the prevailing market direction using a hierarchy of four moving averages. The Fast and Slow EMAs establish the primary trend via their crossover relationship. The Baseline SMA provides a macro filter, and the Hull MA confirms trend transitions with minimal lag. The normalized trend score S_trend ranges from -1.0 (strongly bearish) to +1.0 (strongly bullish).

Setting Name	Type	Default	Range
Fast EMA Length	int	9	Range: 2 - 200
Lookback period for the fast EMA. Lower values (5-8) suit scalping; higher values (12-20) suit swing trading.			
Slow EMA Length	int	21	Range: 2 - 400
Lookback period for the slow EMA. The fast/slow gap defines trend sensitivity. Common pairs: 9/21, 12/26, 20/50.			
Baseline SMA Length	int	50	Range: 5 - 500
Period for the SMA macro filter. Price must be above this SMA for bullish confluence. 50 and 200 are standard institutional levels.			
Hull MA Length	int	14	Range: 2 - 200
Period for the Hull Moving Average. Confirms the EMA crossover direction with reduced phase delay.			
Show Moving Averages	bool	true	Range: on/off
Toggle visibility of all four MA lines on the chart overlay.			

Example: Scalping on ES 5-minute chart

```
Fast EMA Length  = 5
Slow EMA Length  = 13
Baseline SMA      = 34
Hull MA Length   = 9
Rationale: Shorter periods capture quick intraday
reversals on the E-mini S&P 500 futures.
```

Example: Swing trading on daily stock chart

```
Fast EMA Length  = 12
Slow EMA Length  = 26
Baseline SMA      = 200
Hull MA Length   = 20
Rationale: Aligns with the classic MACD relationship
and uses the 200-day SMA institutional benchmark.
```

Section B - Momentum Engine

The Momentum Engine measures the rate and strength of price movement using four oscillators: RSI, MACD, Stochastic, and raw Momentum. Each is normalized to a [-1, +1] scale and averaged into S_momentum. Optional divergence detection flags situations where price and RSI move in opposite directions - a classic reversal warning.

Setting Name	Type	Default	Range
RSI Length	int	14	Range: 2 - 100
Lookback period for the Relative Strength Index. Wilder's original is 14. Shorter (7) = more signals; longer (21) = smoother.			
MACD Fast Length	int	12	Range: 2 - 100
Fast EMA period for MACD calculation. Standard is 12. Decrease for faster response.			
MACD Slow Length	int	26	Range: 2 - 200
Slow EMA period for MACD. Must be greater than MACD Fast. Standard is 26.			
MACD Signal Length	int	9	Range: 2 - 50
EMA period for the MACD signal line. Smooths the MACD line for histogram calculation.			
Stochastic Length	int	14	Range: 2 - 100
Lookback for %K Stochastic oscillator. Maps the close's position within the high-low range.			
Momentum Length	int	10	Range: 1 - 100
Bars for raw momentum (close - close[n]). Measures directional price velocity.			
Enable Divergence	bool	true	Range: on/off
Detect price/RSI divergence. Bullish: price falls while RSI rises. Bearish: price rises while RSI falls.			

Example: Crypto (high-volatility regime)

```

RSI Length      = 10
MACD            = 8 / 21 / 5
Stochastic      = 10
Momentum Length = 7
Divergence      = ON
Rationale: Shorter oscillator periods react faster
to the rapid price swings typical in crypto markets.

```

Section C - Volatility Envelope

The Volatility Envelope measures price dispersion via ATR, Bollinger Bands, and Keltner Channels. A 'squeeze' is detected when the Bollinger Bands contract inside the Keltner Channels, indicating compressed volatility that often precedes a breakout. The volatility percentile rank `S_volatility` indicates whether current volatility is historically high (+1) or low (-1) relative to the last 100 bars.

Setting Name	Type	Default	Range
ATR Length	int	14	Range: 2 - 100
Period for Average True Range. Used for score normalization, stop-loss sizing, and volatility percentile. Wilder's standard is 14.			
BB Length	int	20	Range: 2 - 200
Period for the Bollinger Band middle line (SMA basis). 20 is the standard Bollinger setting.			
BB Multiplier	float	2.0	Range: 0.5 - 5.0
Standard deviation multiplier for BB width. 2.0 captures ~95% of prices. Lower = tighter bands.			
KC Length	int	20	Range: 2 - 200
Period for the Keltner Channel center (EMA basis). Used alongside BB for squeeze detection.			
KC Multiplier	float	1.5	Range: 0.5 - 5.0
ATR multiplier for Keltner Channel width. A squeeze fires when BB contracts inside KC - tune KC mult to adjust sensitivity.			
Show Bollinger Bands	bool	true	Range: on/off
Toggle BB upper/lower bands and fill zone on the chart.			
Show Keltner Channels	bool	false	Range: on/off
Toggle KC upper/lower lines on the chart. Useful for visualizing squeeze conditions.			

Example: Squeeze breakout setup

```
BB Length      = 20, Mult = 2.0
KC Length      = 20, Mult = 1.5
Show BB        = ON
Show KC        = ON

Rationale: With both visible, you can see exactly
when BB contracts inside KC (squeeze = orange dots).
Signals during a squeeze are suppressed by the
volatility confluence check.
```

Section D - Pattern Recognition

The Pattern Recognition Matrix identifies structural price features: EMA crossovers, swing pivot highs/lows, rising/falling bar sequences, and linear regression slope direction. These Boolean flags feed into the confluence gate and visual markers.

Setting Name	Type	Default	Range
Pivot Lookback (Left)	int	5	Range: 1 - 20
Bars to the left that must be lower (for a high) or higher (for a low) to confirm a pivot. Higher = fewer, more reliable pivots.			
Pivot Lookahead (Right)	int	5	Range: 1 - 20
Bars to the right to confirm the pivot. Introduces a lag of this many bars. Trade-off: reliability vs. timeliness.			
Rising/Falling Bars	int	3	Range: 2 - 10
Number of consecutive bars the close must increase (rising) or decrease (falling) to flag directional persistence.			
Linear Regression Len	int	20	Range: 5 - 100
Period for the least-squares regression line. The slope direction (up/down vs. previous bar) indicates statistical trend.			
Show Pivot Points	bool	true	Range: on/off
Display diamond-shaped markers at confirmed swing highs (red-orange) and swing lows (green).			

Example: Higher-timeframe pivots on weekly chart

```
Pivot Lookback    = 10
Pivot Lookahead   = 10
Rising/Falling    = 5
LinReg Length     = 40
Rationale: Wider pivot windows identify major swing
points on weekly charts. Longer rising/falling
requirement filters noise from multi-week trends.
```

Section E - Adaptive Weighting & Confluence

This section controls how the three sub-scores (Trend, Momentum, Volatility) are combined into a single composite score. Weights are auto-normalized to sum to 1.0. The adaptive smoothing option lengthens the EMA smoothing period when volatility is high, making the score less reactive during choppy conditions. The confluence gate requires a minimum number of sub-systems to agree before a signal can fire.

Setting Name	Type	Default	Range
Weight: Trend	float	0.40	Range: 0.0 - 1.0
Relative importance of the Trend Core in the composite score. Auto-normalized with the other two weights.			
Weight: Momentum	float	0.35	Range: 0.0 - 1.0
Relative importance of the Momentum Engine. Increase for mean-reversion strategies; decrease for trend-following.			
Weight: Volatility	float	0.25	Range: 0.0 - 1.0
Relative importance of the Volatility Envelope. Higher weight makes the strategy more volatility-aware.			
Score Smoothing Length	int	5	Range: 1 - 50
Base EMA period for smoothing the composite score. Higher = smoother score line, more lag. 1 = no smoothing.			
Adaptive Smoothing	bool	true	Range: on/off
When ON, smoothing length increases proportionally with the ATR percentile rank. Effect: slower reactions in volatile markets.			
Min Confluence (1-3)	int	2	Range: 1 - 3
Minimum sub-systems that must agree. 1 = any single system can trigger. 2 = two must agree. 3 = all three must agree (strictest).			

Example: Trend-following configuration

```

Weight: Trend      = 0.60
Weight: Momentum   = 0.30
Weight: Volatility = 0.10
Smoothing Length   = 8
Adaptive Smoothing = ON
Min Confluence     = 2
Rationale: Heavily weights the Trend Core for
strong-trend instruments like index futures.
```

Example: Mean-reversion configuration

```

Weight: Trend      = 0.20
Weight: Momentum   = 0.55
Weight: Volatility = 0.25
Smoothing Length   = 3
Adaptive Smoothing = OFF
Min Confluence     = 1
Rationale: Emphasizes momentum oscillators for
range-bound or mean-reverting instruments.
```

Section F - Signal & Risk Management

This section controls when signals fire and how risk is managed. The bull/bear thresholds set the composite score level required for entry. ATR-based stop-loss and take-profit orders are placed automatically. The session gap filter prevents false signals on the first bar after an overnight or weekend gap in futures. Long Only / Short Only modes restrict direction for accounts that cannot short.

Setting Name	Type	Default	Range
Bull Threshold	float	0.25	Range: 0.01 - 0.99
Composite score must exceed this for a long entry. Higher = fewer, higher-conviction signals. Try 0.15 (loose) to 0.40 (strict).			
Bear Threshold	float	-0.25	Range: -0.99 - -0.01
Composite score must fall below this (negative) for a short entry. Mirror of Bull Threshold for the short side.			
Bar Confirmation	bool	true	Range: on/off
When ON, signals only fire after barstate.isconfirmed = true (bar close). Prevents repainting. Turn OFF only for tick-level testing.			
Enable ATR Stop-Loss	bool	true	Range: on/off
Place a stop-loss order at entry +/- (ATR x SL Multiplier). Disabling removes the stop entirely.			
SL ATR Multiplier	float	2.0	Range: 0.5 - 10.0
ATR units for stop-loss distance. 1.5 = tight stop, 3.0 = wide stop. Adjust based on instrument volatility.			
Enable ATR Take-Profit	bool	true	Range: on/off
Place a take-profit at entry +/- (ATR x TP Multiplier). Combined with SL, forms a bracket order.			
TP ATR Multiplier	float	3.0	Range: 0.5 - 20.0
ATR units for take-profit distance. A TP/SL ratio of 1.5x or higher gives positive expectancy even at <50% win rate.			
Filter Session Gaps	bool	true	Range: on/off
Skip signals when bar time gap > 2x expected bar duration. Essential for futures (ES, NQ, CL) to avoid gap-bar false entries.			
Long Only Mode	bool	false	Range: on/off
Restrict strategy to long entries only. Useful for equities in accounts that cannot short.			
Short Only Mode	bool	false	Range: on/off
Restrict strategy to short entries only. Useful for dedicated hedging or bearish-only setups.			
Enable Alerts	bool	true	Range: on/off
Send TradingView alert() notifications on entries. Uses freq_once_per_bar_close to prevent duplicates.			

Example: Conservative futures setup (ES, 15-min)

```

Bull Threshold    = 0.35
Bear Threshold    = -0.35
Bar Confirmation  = ON
ATR Stop-Loss     = ON, Mult = 2.5
ATR Take-Profit   = ON, Mult = 4.0
Filter Gaps       = ON
Long Only         = OFF
Rationale: Higher thresholds and wider SL/TP suit
the higher noise on intraday futures. TP/SL ratio
of 1.6x provides positive risk/reward.
```

Example: Aggressive crypto setup (BTC, 1H)

```

Bull Threshold    = 0.15
Bear Threshold    = -0.15
```

Bar Confirmation = ON
ATR Stop-Loss = ON, Mult = 1.5
ATR Take-Profit = ON, Mult = 3.0
Filter Gaps = OFF
Rationale: Lower thresholds catch more signals.
Crypto trades 24/7 so gap filter is not needed.
2:1 reward/risk compensates for lower win rate.

Section G - Visual Settings

Controls the appearance of the chart overlay: signal colors, background tinting, entry labels, and the real-time dashboard table. These settings do not affect strategy logic or backtesting results - they are purely cosmetic.

Setting Name	Type	Default	Range
Bull Color	color	#26a69a	Range: any color
Color for bullish signals, background tint, and dashboard highlights. Default is teal/green.			
Bear Color	color	#ef5350	Range: any color
Color for bearish signals, background tint, and dashboard highlights. Default is red.			
Neutral Color	color	#787b86	Range: any color
Color for the neutral/inactive state in the dashboard. Default is gray with 50% transparency.			
Show Signal Background	bool	true	Range: on/off
Tint the chart background during active signal regimes. Green tint = bullish, red tint = bearish.			
Show Entry/Exit Labels	bool	true	Range: on/off
Display triangle markers (up for long, down for short) at strategy entry points on the chart.			
Background Transparency	int	85	Range: 50 - 99
Transparency of the signal background. 85 = subtle hint. 50 = bold overlay. 99 = barely visible.			
Show Dashboard Table	bool	true	Range: on/off
Display the ULPR score dashboard in the top-right corner showing composite score, sub-scores, confluence, squeeze, and regime.			

Example: Clean chart (minimal visual clutter)

```
Show MAS           = OFF  (Section A)
Show BB            = OFF  (Section C)
Show KC            = OFF  (Section C)
Show Pivots        = OFF  (Section D)
Show Background    = ON, Transparency = 92
Show Labels        = ON
Show Dashboard     = ON
Rationale: Hides all overlaid indicators, leaving
only the background tint and entry triangles for
a clean price-action-focused view.
```

Configuration Examples by Asset Class

1. Equity Index Futures (ES, NQ) - 15-Minute Chart

Trend Core: Fast EMA 9, Slow EMA 21, SMA 50, HMA 14. Momentum: RSI 14, MACD 12/26/9, Stochastic 14, MOM 10, Divergence ON. Volatility: ATR 14, BB 20/2.0, KC 20/1.5. Weights: Trend 0.45, Momentum 0.35, Volatility 0.20. Confluence 2/3. Thresholds: Bull 0.30, Bear -0.30. SL 2.0 ATR, TP 3.5 ATR. Gap Filter ON. This configuration balances trend sensitivity with noise filtering suitable for the moderate volatility of equity index futures during regular trading hours.

2. Crude Oil Futures (CL) - 1-Hour Chart

Trend Core: Fast EMA 8, Slow EMA 21, SMA 100, HMA 12. Momentum: RSI 10, MACD 8/21/5, Stochastic 10, MOM 8, Divergence ON. Volatility: ATR 14, BB 20/2.5, KC 20/2.0. Weights: Trend 0.35, Momentum 0.40, Volatility 0.25. Confluence 2/3. Thresholds: Bull 0.20, Bear -0.20. SL 2.5 ATR, TP 4.0 ATR. Gap Filter ON. Higher BB/KC multipliers account for CL's wider price swings. Momentum is weighted more heavily due to CL's tendency for sharp directional moves.

3. Forex Major (EUR/USD) - 4-Hour Chart

Trend Core: Fast EMA 12, Slow EMA 26, SMA 200, HMA 20. Momentum: RSI 14, MACD 12/26/9, Stochastic 14, MOM 14, Divergence ON. Volatility: ATR 14, BB 20/2.0, KC 20/1.5. Weights: Trend 0.50, Momentum 0.30, Volatility 0.20. Confluence 2/3. Thresholds: Bull 0.25, Bear -0.25. SL 1.5 ATR, TP 2.5 ATR. Gap Filter OFF. Forex trades nearly 24 hours so gap filter is unnecessary. The 200-period SMA aligns with the institutional benchmark on the 4H timeframe. Trend is heavily weighted because forex pairs tend to exhibit sustained directional moves.

4. Cryptocurrency (BTC/USD) - 1-Hour Chart

Trend Core: Fast EMA 7, Slow EMA 18, SMA 50, HMA 10. Momentum: RSI 10, MACD 8/21/5, Stochastic 10, MOM 7, Divergence ON. Volatility: ATR 14, BB 20/2.5, KC 20/2.0. Weights: Trend 0.30, Momentum 0.45, Volatility 0.25. Confluence 2/3. Thresholds: Bull 0.15, Bear -0.15. SL 1.5 ATR, TP 3.0 ATR. Gap Filter OFF. Crypto markets are 24/7 with high volatility. Shorter periods and lower thresholds capture the frequent momentum-driven moves. Wider BB/KC multipliers prevent excessive squeeze signals. Momentum is the dominant weight.

5. Stocks (Daily Chart, Long Only)

Trend Core: Fast EMA 10, Slow EMA 30, SMA 200, HMA 16. Momentum: RSI 14, MACD 12/26/9, Stochastic 14, MOM 14, Divergence ON. Volatility: ATR 14, BB 20/2.0, KC 20/1.5. Weights: Trend 0.55, Momentum 0.30, Volatility 0.15. Confluence 2/3. Thresholds: Bull 0.30. SL 2.0 ATR, TP 4.0 ATR. Long Only Mode ON. For buy-and-hold-oriented stock accounts. The 200-day SMA filters for secular uptrends. Long Only mode prevents short positions. Higher TP/SL ratio (2:1) compensates for the asymmetric upward bias in equities.

Appendix - Formula Quick Reference

Formula	Definition
Trend Score (S_trend)	clamp((EMA_fast - EMA_slow) / ATR , -1, +1)
Momentum Score (S_momentum)	avg((RSI-50)/50, clamp(MACD_hist/ATR), (Stoch-50)/50, clamp(MOM/(2*ATR)))
Volatility Score (S_volatility)	(percentrank(ATR, 100) - 50) / 50
Composite Score	alpha * rawScore + (1-alpha) * prevScore, where rawScore = w1*S_trend + w2*S_mom + w3*S_vol
Adaptive Alpha	2 / (dynamicLen + 1), where dynamicLen = smoothLen * (1 + volPctRank/100)
Stop-Loss Distance	ATR * SL_Multiplier (in price units from entry)
Take-Profit Distance	ATR * TP_Multiplier (in price units from entry)
Squeeze Condition	BB_upper < KC_upper AND BB_lower > KC_lower
Bull Divergence	ta.change(close, 5) < 0 AND ta.change(RSI, 5) > 0
Bear Divergence	ta.change(close, 5) > 0 AND ta.change(RSI, 5) < 0
Session Gap	ta.change(time) > 2 * timeframe.in_seconds() * 1000
Long Entry Condition	compositeScore > bullThreshold AND confluence >= minConfluence AND confirmed AND NOT gap
Short Entry Condition	compositeScore < bearThreshold AND confluence >= minConfluence AND confirmed AND NOT gap

Note: clamp(x) = math.max(-1, math.min(1, x)). All PineScript v6 functions (ta.ema, ta.sma, ta.hma, ta.rsi, ta.macd, ta.stoch, ta.mom, ta.atr, ta.bb, ta.kc, ta.crossover, ta.crossunder, ta.pivohigh, ta.pivotlow, ta.rising, ta.falling, ta.linreg, ta.change, ta.percentrank) are documented in the TradingView PineScript v6 Language Reference Manual.